MCAIT 2026 talk script

Fakhrul Azim · Parallel Session 6 · Plain-language version for the 25-slide deck (179.pptx)

> **How to use it.** Read it out loud in your own words. The wording is plain on purpose, so it sounds like talking, not reading.
>
> **The one sentence to remember:** an interface can only honestly show what the search behind it can back up.
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> Gold italic lines are cues, not words to say.

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1 Title

Good afternoon, everyone. I am Fakhrul Azim, from Universiti Teknologi PETRONAS in Malaysia. My paper starts with a simple question. When you study from many of your own documents, why do you still just get a chat box? And what would it take to give you something better?

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2 The claim of this paper

Say the sentence slowly.

Here is the whole paper in one sentence. An interface can only honestly show what the search behind it can actually find. If the search cannot find how two sources connect, the screen should not claim that they do. I will also show Arcana, which is this paper in practice, and some results we measured after the paper was accepted.

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3 Learners ask relational questions and get prose

When you study across many documents, the useful questions are about how things connect. Which papers agree? Where do they disagree? What is missing? Today's tools answer with a paragraph, so you do the connecting yourself. Interactive visuals can help a lot. One review of twenty physics studies found a large effect, 0.83. But people built those by hand. Tools for your own documents rarely offer them. Someone has to build each one for each topic, and the search has nothing organised to fill it with.

4 Flat retrieval has no way to represent a relation

The cause is in the search. Normal search scores each passage on its own. It finds passages close to your question, but keeps no record that two passages disagree, or that one idea depends on another. A bigger model does not fix this, because the problem is what is stored. A knowledge graph stores the ideas and the links between them. So if two documents mention the same idea, like temperature here, the search can bring both passages together.

5 Four literatures, and the gap between them

Four areas of earlier work touch this. Graph search is judged on answer quality, not on what it lets you display. Generated interfaces are judged on whether people like them, which is not the same as being correct. Education tools keep a teacher in the loop, which is safe but limited to what the teacher planned. And checking methods rely on test suites, which a lesson does not have. One recent paper argues that interfaces written on the fly cannot be checked at scale. In what I reviewed, nothing answers that. That is the gap.

6 Generative interfaces are reaching the market

This is not only a research question. Gemini Notebook, which used to be NotebookLM, lets you pick an output like a mind map or a table. ChatGPT's Intelligent UI picks a format for you. Gemini's dynamic view writes custom code for each prompt. I only report what the companies publish, and I have not tested these products. None of those descriptions states a rule linking what is shown to the structure that supports it. Stating that rule is what this paper does.

7 Contributions

The paper makes four contributions. First, a design that joins graph search to an interface built from fixed, checked parts. Second, an interface put together for each question, with checks before anything reaches you. Third, a rule showing that what the search can find limits what the interface can show. Fourth, a test plan written before any results, with the pilot reported exactly as it came out. Arcana is how these ideas look in practice. The claims belong to the paper.

8 The method in thirty seconds

Click the video as you start talking. About 29 seconds.

Here is the whole method in thirty seconds. You ask a question that compares things, and today you get a paragraph. Instead, the documents become ideas and links, so sources connect. Three kinds of search are combined, so evidence from different documents arrives together. Then the model builds the answer from approved parts. It writes no code, and a faulty part is rejected at the gate. And what the search can find sets the limit. Basic search can fill three of these parts. Graph search can fill all six.

9 Three regimes for producing a learning interface

There are three ways to make a learning interface. Prepare it ahead: safe, but it only covers what someone planned. Let the model write interface code live: it can cover anything, but it is very hard to check. Or what we do: the model only chooses from approved parts and fills them with evidence. That keeps the wide coverage and keeps it accountable. The price is flexibility. If the best answer would be a view we have not built, you get the closest one we have.

10 From document to rendered component

Here is the design in three steps. First, documents are split into passages, and in the same pass we pull out the ideas and links, with no fixed list of topics. Second, when you ask, three searches run: by meaning, by keywords, and by following graph links. We combine them by rank, because their scores cannot be compared. Passages ranked high, or found by more than one search, rise to the top. Third, the model picks parts and fills them from that evidence, and everything is checked first.

11 The model composes the interface. It does not write it.

This is what makes it trustworthy. The model does not write the interface. It picks from fixed parts, and the content comes from what was found. Each safeguard in this table stops one specific failure. The check runs on the server and rejects anything wrong. We prefer showing an error to showing a doubtful answer, because you can recover from an error, but not from a confident wrong answer. The paper also states two gaps. The claim check lets content through if the checking service fails, and contradiction quotes are not yet matched word for word.

12 Adaptive interface: where expressiveness shows

Click the video as you start talking. About 17 seconds.

This is where the flexibility shows. There are very many possible interfaces: which parts, in what order, and what goes inside each. So the screen adapts. A question about one paper gets an answer and sources. A comparison gets a dashboard. If the links connect, a concept map appears. The workspace also changes with the task, including studying. Nobody designed these layouts in advance.

13 A worked request from the paper

Here is the example from the paper. A student revising chemistry asks why a higher temperature makes a reaction faster. Their notes say particles collide more often. The textbook says mainly that more collisions have enough energy. Both are true, but that difference in emphasis can cost marks in an exam. The graph brings both passages together, and the model spots the disagreement. The answer is built from three parts nobody prepared for this question. A biology question would get a comparison grid instead. This is an illustration from the paper, not a recording.

14 Retrieval structure bounds interface expressiveness

The most important slide. Slow down.

This is the central idea. Every part of an interface makes a claim about the shape of the answer. A comparison grid claims documents line up on shared points. A concept map claims ideas are linked in specific ways. So we ask which parts the search can back up with evidence it actually found. Basic search can back up a summary and a source list. It cannot back up a map of links, because it never retrieved any. A model can still draw one, but the links would come from the model, not the evidence. I should say that graph search covering everything basic search does, and more, is true by definition. Its value is as a design rule.

15 Committed protocol, and a pilot reported as found

Now the testing. We wrote the test plan before running anything, and fixed the twenty questions on 9 June 2026. A provider limit stopped our extraction partway, so the paper reports a pilot on four of sixteen documents. The result was a tie. Basic search and graph-plus-basic search matched on eighteen of twenty questions. The reason is the finding. The graph search found nothing on fifteen questions, because the question's words had to match a graph entry exactly. So the pilot did not really test the graph. It showed that the limit is matching questions to the right entries.

16 Where Arcana began: VERA AI

A word on where this came from. Arcana grew out of VERA AI, a company knowledge assistant from my internship. VERA already searched well and would not answer without a citation. But documents sat in separate silos, the layout was fixed, and the agents did not work together. It could cite its answers, but had nothing to show how sources relate. That is the limit this paper explains. This compares designs only, not results, because the domains and documents are different.

17 Arcana: the paper in practice

Click the video as you start talking. About 36 seconds.

This is Arcana on the sixteen retrieval papers. First, three questions with a real model. A question about connections gets a cited summary and a graph. A comparison gets a grid. A question the papers cannot answer gets a refusal. Next, the learning tools. Concept pages show the sentences behind each link. A wrong quiz answer gets corrected, and the graph changes colour by how well you know each idea. And a lab appears only where a passage states the formula.

18 What retrieval structure makes possible

Here is what that structure looks like. On the left is part of the real graph from the sixteen papers, showing ideas that appear in the most papers. About one in nine ideas appears in two or more papers, and those are where documents connect. You can also see a weakness: "LLM" and "LLMs" are separate dots. On the right is part of a comparison grid built from that graph. Each cell has a quote and a page number. If a paper has nothing on the topic, the cell says so, instead of making something up.

19 Beyond the catalogue: generated, then verified

The paper admits a cost: the menu of parts is a ceiling. If the best answer is an interactive experiment, you get the closest part. Our team's system, Synapse, explores a way past that for school science. The model plans a lab from a checked set of building blocks, and a program runs it in a real browser before any student sees it. In tests, it refused 219 of 224 faulty labs we planted. We wrote those faults ourselves, so that is a regression check, not a general rate. This is a different area, so I am not comparing results. Synapse is by Team EduNova and is not part of the paper.

20 On the full corpus, a retrieval gain is not shown

This is the slide that earns trust. Say it plainly.

Now the results since the paper. We finished the study the paper called for: all sixteen papers, fixed matching, and approved right passages for each question. On the original, harder questions that span documents, the combined search is a little ahead, but not reliably. On nineteen new questions, which lean toward single papers, it falls behind. Put together, the difference is basically zero. So I do not claim a search improvement. Not shown to help is not the same as shown not to help, but this data does not back the claim. This is a different score from the pilot, so please do not compare them.

21 The graph does extend the interface: 2 components against 5

Here is the other half. For each question we counted which parts had enough evidence to be shown honestly. Basic search qualifies for two. Graph search qualifies for all five. Basic search cannot fill the other three because of how we defined them, so that is not a discovery. What we measured is how often the evidence exists: a link diagram on sixteen of twenty questions, but a multi-step path on only three, because every step needs a sentence behind it. And qualifying does not mean useful. The grid qualified on every question, so it is not yet selective.

22 Grounding holds with a real model, relevance is not yet shown

Do not rush the second row.

We also ran the full system with a real model. All thirty-eight answerable questions were answered, and the one unanswerable question was refused. All 125 quotes matched the source word for word. And the checker removed ten claims that had a real quote attached but were not supported by it. Now the second row. Only about 21 percent of cited passages were on our approved list. That list is incomplete, so this does not prove the answers are wrong. But it shows that being tied to a source and being the most useful source are different things. This was one run, with one model.

23 What this does and does not show

Slow down for the last sentence, then pause.

To finish. What is established: the interface rules hold by design, and the limit follows from how we defined the parts. What is not shown: any search improvement from the graph on this data, and any learning benefit, because we have not run a user study. What comes next: a human check of the graph's links, and the user study itself. If you remember one thing, let it be this. An interface can only honestly show what the search behind it can back up.

24 Selected references

These are the references used on the slides, numbered as in the paper. The full list is in the proceedings.

25 Thank you

Thank you. The code and the test plan are public, and I am happy to take questions.